

NANDAKUMAR VUPPALAPATI

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PROFESSIONAL SUMMARY

Senior Data Engineer with 4+ years designing and scaling production ETL/ELT pipelines and Lakehouse platforms across AWS, Azure, and Databricks. Specialized in high throughput streaming and batch architectures (Spark, Kafka, dbt, Airflow) for regulated financial and healthcare environments, with a proven record in dimensional modeling, data contract enforcement, cloud cost optimization, and enterprise RAG pipelines on AWS Bedrock.

TECHNICAL SKILLS

- **Data Processing & Streaming:** Apache Spark (PySpark Engine), Databricks Delta Lake, Spark Structured Streaming, Apache Kafka, Apache Flink
- **Orchestration & Transformation:** Apache Airflow (DAG Design, Sensors, Task Groups), dbt (Core/Cloud), Azure Data Factory, AWS Step Functions
- **Data Warehousing & Lakehouse:** Snowflake, Databricks, AWS Redshift, Azure Synapse, Medallion Architecture (Bronze/Silver/Gold)
- **Cloud & DevOps:** AWS (S3, Glue, Lambda, Bedrock, Redshift, Athena), Azure (ADLS Gen2, Data Factory, Synapse), Terraform, Docker, GitHub Actions (CI/CD)
- **Data Governance & Quality:** Great Expectations, Data Contracts, Data Lineage, PII Anonymization, SOX-Aware Controls, RBAC, Unity Catalog
- **Databases & AI Systems:** PostgreSQL, Databricks SQL, Pinecone, MLflow, AWS Bedrock, Feast Feature Store, FastAPI, Advanced SQL Analytics

WORK EXPERIENCE

Workiva | Senior *Data Engineer*

Remote, USA | Dec 2025 – Present

- Architected multi-tenant, high-throughput pipeline infrastructure using Apache Spark, Airflow, and AWS (Glue, S3, Lambda, Step Functions) to process 1M+ daily financial records, maintaining 99.9% data SLA compliance for risk and settlement analytics.
- Engineered modular dbt transformation frameworks on Snowflake with custom macros and data contracts, centralizing 30+ core financial data models to support executive reporting and strict SOX compliance.
- Optimized distributed PySpark batch jobs and Snowflake warehouse sizing, driving a 25% reduction in monthly cloud compute overhead while accelerating dashboard query execution speeds.
- Implemented continuous data observability by integrating Prometheus and Grafana alerting into Airflow orchestration loops, reducing Mean-Time-To-Detection (MTTD) for pipeline anomalies by 60%.
- Architected an enterprise regulatory search pipeline using AWS Bedrock and Pinecone vector databases, enabling sub-second natural language querying across thousands of SEC compliance filings.

Wellmark Blue Cross Blue Shield | *Data Engineer*

Des Moines, IA | Aug 2024 – May 2025

- Engineered Azure Data Factory ingestion pipelines processing 10M+ records from 30+ legacy healthcare sources into Azure ADLS Gen2 and Azure SQL, applying automated PII masking and zero-trust security controls.
- Modernized legacy SQL processing into scalable Azure Databricks PySpark jobs across Medallion architecture (Bronze/Silver/Gold), increasing data ingestion throughput by 45%.
- Enforced Great Expectations data quality checkpoints and dbt schema assertions across ingestion boundaries, preventing upstream schema drift from breaking downstream reporting pipelines and eliminating emergency on-call interventions.
- Provisioned cloud data infrastructure using Terraform, Docker, and GitHub Actions CI/CD pipelines, automating multi-environment deployments across staging and production.

Sigmoid | *Data Engineer*

Bengaluru, India | Jun 2022 – Dec 2023

- Refactored mission-critical T-SQL queries and indexing strategies in Azure Synapse analytics pools, cutting month-end financial reporting query times from 4 hours to under 15 minutes.
- Redesigned legacy flat reporting structures into an optimized star-schema dimensional model in Azure Synapse, consolidating enterprise financial metrics into a unified warehouse for multi-region reporting.
- Engineered an automated data validation and reconciliation framework using Python, PyTest, and Delta Lake assertions, reducing upstream data corruption incidents by 70% and ensuring audit reliability.
- Managed production pipeline reliability in Azure Data Factory across 40+ concurrent jobs, implementing automated retries, failure hooks, and dead-letter queue routing to guarantee uninterrupted 24/7 availability.

Tech Citi Software | Associate *Data Engineer*

Bengaluru, India | Jul 2021 – Dec 2021

- Migrated historical financial transaction data from Oracle to PostgreSQL, executing schema mapping, datatype alignment, and automated reconciliation validation.
- Assembled Apache Airflow ETL workflows to ingest 50,000+ daily financial transaction records via REST APIs and flat files, replacing legacy manual data-loading processes.

KEY PROJECTS

Real-Time ML Feature Platform for Fraud Scoring | [GitHub](#) | *Kafka, Flink, Feast, Databricks, AWS*

- Streaming feature engineering pipeline using Kafka and Flink, serving low-latency features via Feast and FastAPI to downstream ML models.

CDC Lakehouse with Data Contracts | [GitHub](#) | *Debezium, Kafka, Apache Iceberg, dbt, Docker*

- Change-data-capture lakehouse pipeline to Apache Iceberg, enforcing schema integrity via dbt contracts to auto-quarantine malformed records.

RAG Evaluation & Observability Harness | [GitHub](#) | *LangChain, OpenAI API, Pinecone, Ragas, FastAPI, Airflow*

- Regression-testing harness separating RAG failures into retrieval vs. generation issues, scoring golden-question datasets against SEC filings with Ragas on Airflow.

EDUCATION & CERTIFICATIONS

- Masters in computer and information science — University of South Dakota (2025)
- Bachelor of Technology — IIIT, India (2023)
- AWS Certified Cloud Practitioner (2025)